

LSM Semester Exam Papers

Supp Midsem P1

Question 1.

Total marks: 18

Consider the linear regression model

$$y_i = \alpha + \beta x_i + \gamma z_i + \epsilon_i, \quad i = 1, \dots, n,$$

where $x_1, \dots, x_n, z_1, \dots, z_n$ are fixed numbers, and $\epsilon_1, \dots, \epsilon_n \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$.

- (a) Find the least-squares estimators $\hat{\alpha}_{\text{LSE}}$, $\hat{\beta}_{\text{LSE}}$ and $\hat{\gamma}_{\text{LSE}}$ of α , β and γ . [5 marks]
- (b) Find $\text{Var}(\hat{\alpha}_{\text{LSE}})$, $\text{Var}(\hat{\beta}_{\text{LSE}})$, $\text{Var}(\hat{\gamma}_{\text{LSE}})$ and $\text{Cov}(\hat{\beta}_{\text{LSE}}, \hat{\gamma}_{\text{LSE}})$. [8 marks]
- (c) Formulate a test for $H_0 : \beta = \gamma$ against $H_a : \beta \neq \gamma$. Clearly define the test statistic, the cut-off and the rejection rule. Justify your choice of test statistic. [5 marks]

Supp Midsem P2

Question 1.

Total marks: 20

Let $\{(x_1, y_1), \dots, (x_n, y_n)\}$ be n pairs of observations. We assume the linear regression model

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where $\epsilon_1, \dots, \epsilon_n \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$.

- (a) Derive the least-squares estimators $\hat{\alpha}$ and $\hat{\beta}$ of α and β . [2 marks]
- (b) Derive the joint distribution of $\hat{\alpha}$ and $\hat{\beta}$. [5 marks]
- (c) We want to test $H_0 : \alpha + \beta = 5$ against $H_a : \alpha + \beta \neq 5$. Formulate a test for this problem. Clearly specify the test statistic, the cut-off and the rejection rule. [5 marks]
- (d) The following information is given to you:

$$n = 50, \quad \sum_{i=1}^{50} x_i = 20, \quad \sum_{i=1}^{50} y_i = 30, \quad \sum_{i=1}^{50} x_i^2 = 40, \quad \sum_{i=1}^{50} y_i^2 = 36, \quad \sum_{i=1}^{50} x_i y_i = 10.$$

Calculate $\text{Var}(\hat{\alpha})$, $\text{Var}(\hat{\beta})$ and $\text{Cov}(\hat{\alpha}, \hat{\beta})$. Also calculate the value of the test statistic in part (c). [8 marks]

Midsem P1

Instructions

- There are two papers in the mid-semester exam: Paper 1 and Paper 2.
- Answer as much as you can. The maximum you can score is 30 (in Papers 1 & 2 combined).
- You can use any result obtained in the class. Clearly state the results that you use.
- Clearly state all the assumptions that you make. Failure to do so will lead to deduction of marks.
- Keep your answers to the point. Any unnecessary or irrelevant writing will be penalized.

Question 1.

Total marks: 18

Consider the linear regression model

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where $x_1, \dots, x_n$ are fixed numbers with $x_i \neq x_j$ for some $i \neq j$, and $\epsilon_1, \dots, \epsilon_n$ are i.i.d. with $E(\epsilon_i) = 0$ and $\text{Var}(\epsilon_i) = \sigma^2$.

- (a) Find the least-squares estimators $(\hat{\alpha}_{\text{LSE}}, \hat{\beta}_{\text{LSE}})$ of (α, β) . Derive $\text{Var}(\hat{\alpha}_{\text{LSE}})$ and $\text{Var}(\hat{\beta}_{\text{LSE}})$. [2+3 marks]
- (b) State the Gauss–Markov theorem in the context of the linear regression model given above. [2 marks]
- (c) Now, consider the following estimators of α and β :

$$(\hat{\alpha}_\lambda, \hat{\beta}_\lambda) = \arg \min_{a,b} \frac{1}{n} \sum_{i=1}^n (y_i - a - bx_i)^2 + \lambda b^2,$$

where $\lambda > 0$ is a fixed (known) number. Find the expressions for $\hat{\alpha}_\lambda$ and $\hat{\beta}_\lambda$. [3 marks]

- (d) Derive $\text{Var}(\hat{\alpha}_\lambda)$ and $\text{Var}(\hat{\beta}_\lambda)$. [3 marks]
- (e) Show that $\text{Var}(\hat{\alpha}_\lambda) < \text{Var}(\hat{\alpha}_{\text{LSE}})$ and $\text{Var}(\hat{\beta}_\lambda) < \text{Var}(\hat{\beta}_{\text{LSE}})$. Is this a violation of the Gauss–Markov theorem? Explain. [2+3 marks]

Midsem P2

Instructions

- There are two papers in the mid-semester exam: Paper 1 and Paper 2.
- Answer as much as you can. The maximum you can score is 30 (in Papers 1 & 2 combined).
- You can use any result obtained in the class. Clearly state the results that you use.
- Clearly state all the assumptions that you make. Failure to do so will lead to deduction of marks.
- Keep your answers to the point. Any unnecessary or irrelevant writing will be penalized.

Question 1.

Total marks: 18

Let $\{(x_1, y_1), \dots, (x_n, y_n)\}$ be n pairs of observations. We assume the linear regression model

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where $\epsilon_1, \dots, \epsilon_n \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. Consider the problem of testing $H_0 : \beta = 0$ vs. $H_a : \beta \neq 0$.

- (a) Derive the F -test for the testing problem above. Clearly specify the test statistic, the cut-off value, and the rejection rule. [6 marks]
- (b) Show that the F -test statistic for this problem can be written as a function of the sample correlation coefficient between x and y . [8 marks]
- (c) The following information is given to you:

$$n = 50, \quad \sum_{i=1}^{50} x_i = 20, \quad \sum_{i=1}^{50} y_i = 30, \quad \sum_{i=1}^{50} x_i^2 = 40, \quad \sum_{i=1}^{50} y_i^2 = 36, \quad \sum_{i=1}^{50} x_i y_i = 10.$$

Calculate the value of the F -test statistic. [4 marks]

Endsem P1

Instructions

- There are two papers in the semester exam: Paper 1 and Paper 2.
- Answer as much as you can. The maximum you can score is 50 (in Papers 1 & 2 combined).
- You can use any result obtained in the class. Clearly state the results that you use.
- Clearly state all the assumptions that you make. Failure to do so will lead to deduction of marks.
- Keep your answers to the point. Any unnecessary or irrelevant writing will be penalized.

Question 1.

Total marks: 15

A food enthusiast wants to study the effect of two different flour varieties on the quality of pizzas. For this study, he gathers one pack of each flour variety. Now, each pack of flour is sufficient to create three pizza doughs, so he decides to make three different types of pizzas (Margherita, Napoletana, and Tropicana) with each of the two packs of flour. After preparing the pizzas, he measures the thickness of the crust and models it as

$$\text{thickness} = \mu + f_i + p_j + \text{error},$$

if flour variety i is used to prepare pizza type j .

Here, f_i is the effect of the i -th flour variety and p_j is the effect of the j -th pizza type. The error part is assumed to be i.i.d. $N(0, \sigma^2)$.

- (a) Express the model in vector-matrix notation $y = X\theta + \epsilon$. Clearly define all the components of the model. [2+1 marks]
- (b) Show that f_1, f_2 are not estimable from the model, but $f_1 - f_2$ is estimable. [3 marks]
- (c) Find, with justification, the best linear unbiased estimator of $f_1 - f_2$ from the model. Also find its distribution. [4 marks]
- (d) Formulate a statistical test for checking whether the effect of the two flour varieties on the thickness of the pizzas are different. Clearly define the hypothesis, the test statistic, the critical value, and the rejection rule. Justify the choice of test statistic. [5 marks]

Question 2.**Total marks: 15**

We have three sets of observations D_1 , D_2 and D_3 , where

$$D_j = \{(x_{j1}, y_{j1}), (x_{j2}, y_{j2}), \dots, (x_{jm}, y_{jm})\}, \quad j = 1, 2, 3.$$

We are interested in the relationship between x and y for these three sets of data.

- (a) We believe that y is an approximately linear function in x for each of the three datasets. Formulate a linear model that reflects our belief. Find estimators of the slopes and intercepts of the regression lines in the three datasets. [2+3 marks]
- (b) We want to check whether the three lines in part (a) are parallel to each other. Formulate this as a statistical hypothesis testing problem. Construct an appropriate test for this problem, clearly defining the test statistic, the critical value and the rejection rule. [5 marks]
- (c) You are given the following information:

Dataset	S_x	S_y	S_{xx}	S_{yy}	S_{xy}
D_1	1.67	90.52	54.34	166.72	14.06
D_2	4.45	23.31	54.60	28.91	32.88
D_3	8.78	-30.09	46.43	34.44	21.29

Here, $S_x = \sum_{i=1}^m x_{ij}$, $S_y = \sum_{i=1}^m y_{ij}$, $S_{xx} = \sum_{i=1}^m x_{ij}^2$, $S_{yy} = \sum_{i=1}^m y_{ij}^2$, and $S_{xy} = \sum_{i=1}^m x_{ij}y_{ij}$ for $j = 1, 2, 3$. You are also told that $m = 50$. Compute the estimates of the intercepts and slopes in part (a). Also calculate the value of the test statistic in part (b). [3+2 marks]

Endsem P2

Instructions

- There are two papers in the semester exam: Paper 1 and Paper 2.
- Answer as much as you can. The maximum you can score is 50 (in Papers 1 & 2 combined).
- You can use any result obtained in the class. Clearly state the results that you use.
- Clearly state all the assumptions that you make. Failure to do so will lead to deduction of marks.
- Keep your answers to the point. Any unnecessary or irrelevant writing will be penalized.

Question 1.

Total marks: 12

In order to compare the effectiveness of three drugs in controlling the blood sugar levels of patients, the following model is used

$$y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, 2, 3, \quad j = 1, \dots, n_i,$$

where y_{ij} is the change in the blood sugar level of the j -th patient receiving drug i , μ is the overall effect, τ_i is the additional effect due to drug i , and ϵ_{ij} is the error component. It is further assumed that $\epsilon_{ij} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$.

- (a) Show that the hypothesis $H_0 : \tau_1 = \tau_2 = \tau_3$ is testable. [2 marks]
- (b) Construct the F -test statistic for testing H_0 against the general alternative $H_a : H_0$ is not true. Clearly define the test statistic, the critical value and the rejection rule. [5 marks]
- (c) Suppose that $n_1 = 52$, $n_2 = 37$, $n_3 = 148$, $\sum_{j=1}^{52} y_{1j} = 264.37$, $\sum_{j=1}^{37} y_{2j} = 99.24$, and $\sum_{j=1}^{148} y_{3j} = 606.18$, and $\sum_{i=1}^3 \sum_{j=1}^{n_i} y_{ij}^2 = 4316.19$. Based on this information, construct the one-way ANOVA table for the given problem. Find the value of the F -test statistic in part (b). [3+2 marks]

Question 2.**Total marks: 18**

A food delivery company delivers food from 50 restaurants to 90 locations (pincodes). In order to study the variation in the order values (amount spent on one particular order), the company considers the orders from each (restaurant, location) pair in the last month, notes the averages of the corresponding orders and uses the following two-way random effects model:

$$ov_{ij} = \mu + r_i + l_j + \epsilon_{ij}, \quad i = 1, \dots, 50, \quad j = 1, \dots, 90,$$

where ov_{ij} is the average order value from the i -th restaurant and the j -th location, μ is the overall effect, r_i is the effect due to the i -th restaurant, l_j is the effect due to the j -th location, and ϵ_{ij} is the error component. Also, the following assumptions are made:

$$r_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_r^2), \quad l_j \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_l^2), \quad \epsilon_{ij} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_e^2),$$

and all are independent.

- (a) Prove or disprove: The observations ov_{ij} are independent in the above model. [2 marks]

- (b) Let

$$SS_r = 90 \sum_{i=1}^{50} (ov_{i.} - ov_{..})^2, \quad SS_l = 50 \sum_{j=1}^{90} (ov_{.j} - ov_{..})^2,$$

and

$$SS_e = \sum_{i=1}^{50} \sum_{j=1}^{90} (ov_{ij} - ov_{i.} - ov_{.j} + ov_{..})^2,$$

where

$$ov_{i.} = \frac{1}{90} \sum_{j=1}^{90} ov_{ij}, \quad ov_{.j} = \frac{1}{50} \sum_{i=1}^{50} ov_{ij}, \quad ov_{..} = \frac{1}{50} \sum_{i=1}^{50} ov_{i.} = \frac{1}{90} \sum_{j=1}^{90} ov_{.j}$$

Show that SS_r , SS_l and SS_e are independent, and they follow scaled χ^2 distribution, i.e. they are of the form $c\chi_\nu^2$ for some c and ν . Identify the parameters c and ν in each case. [12 marks]

- (c) The company wants to know whether the location effect on the order value is significant. Formulate this as a statistical hypothesis testing problem under the above setup. Construct a test for this problem. Clearly specify the test statistic, the critical value and the rejection rule. [4 marks]